Cybersickness Abatement From Repeated Exposure Generalizes Across Experiences

Jonathan W. Kelly, Taylor A. Doty, Michael C. Dorneich, and Stephen B. Gilbert

***Abstract*—Cybersickness, or sickness caused by virtual reality (VR), represents a significant threat to the usability of VR applications. Repeated exposure to the same VR stimulus causes a reduction in cybersickness, referred to as Cybersickness Abatement from Repeated Exposure (CARE). This study examined whether the benefits of CARE generalize across distinct VR contexts, which was operationalized as three distinct games (a climbing game, a puzzle game, and a stealth survival game). Participants played a VR game for up to 20 minutes. Those in the Repeated Exposure condition played one VR game (either a puzzle game or a climbing game) on three separate days followed by a different VR game (a survival game) on the fourth day. Those in the Single Exposure condition played the survival game once. The three games all differed in several ways, including environment and task, whereas the puzzle and survival games shared a similar joystick locomotion interface that differed from the locomotion interface in the climbing game. Results indicate that cybersickness on Day 4 of the Repeated Exposure condition was significantly lower than that in the Single Exposure condition, regardless of which game was experienced on Days 1-3. The practical implication of this finding is that CARE that occurs in one VR context can generalize to a novel context with a distinct environment, task, and locomotion interface. Results are considered in the context of multiple theoretical explanations for CARE, including sensory rearrangement and habituation. These results support systematic exposure as an approach to reducing cybersickness.**

***Index Terms*—Cybersickness, virtual reality, motion sickness, sensory conflict, sensory rearrangement, adaptation, habituation.**

I. Introduction

CYBERSICKNESS, or discomfort caused by virtual reality (VR), commonly includes symptoms such as nausea, headache, eye strain, and disorientation [1], [2], [3], [4]. Cybersickness, also sometimes called virtual reality induced symptoms and effects (VRISE) [5], can affect up to half of VR users in just 10-20 minutes [6], [7] and presents a significant barrier to VR usability. Some individuals are at increased risk for cybersickness, including women [8], [9] and people with a history of motion sickness [10], [11], [12].

Received 12 February 2025; revised 26 January 2026; accepted 28 January 2026. Date of publication 2 February 2026; date of current version 2 March 2026. This work was supported by the National Science Foundation under Grant 2309990. Recommended for acceptance by J. P. Quarles. *(Corresponding author: Jonathan W. Kelly.)*

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Digital Object Identifier 10.1109/TVCG.2026.3659810

Repeated exposure to VR has been found to reduce cybersickness [13], [14], [15], [16], [17], [18], [19], a finding referred to as Cybersickness Abatement from Repeated Exposure (CARE) [14], a term that describes the effect but remains neutral with respect to possible underlying mechanisms (e.g., habituation, adaptation, task familiarization, etc.). With sufficient supporting research, CARE could become a useful tool to combat cybersickness through systematic exposure to VR. Ideally, CARE that occurs in one context (e.g., a driving scenario) would generalize to another context (e.g., a flying scenario). CARE can involve initial discomfort until the benefits of repeated exposure accrue, so generalization of CARE across VR contexts (environments, interfaces, tasks, etc.) would reduce the initial discomfort that might otherwise occur when beginning a new VR simulation.

The current study examined whether CARE generalizes across VR contexts and whether the degree of generalization depends on similarity of the locomotion interface between exposure and novel contexts. Specifically, we asked whether CARE acquired in one VR context transfers to a distinct context that differs in task, environment, and interface. Sensory rearrangement accounts predict greater generalization when the locomotion interface is similar across contexts, whereas habituation accounts predict comparable generalization regardless of interface similarity. These predictions were evaluated by comparing cybersickness following prior exposure to a novel VR experience with cybersickness observed in participants without prior exposure.

II. Related Work

A. Cybersickness Abatement From Repeated Exposure (CARE)

It is now well established that repeated exposure to the same VR experience causes reduction in cybersickness [13], [14], [15], [16], [17], [18], [19], [20], [21], or CARE. In a prototypical study [21], participants played a racing game on a VR headset for up to 20 minutes or until they could no longer continue due to sickness. Sickness ratings were taken after each exposure and every minute during exposure. Participants experienced 10 VR exposures on separate days. Sickness scores obtained on Day 10 were significantly lower than those on Day 1, indicating that CARE occurred. CARE has subsequently been identified as early as the second VR exposure [16], [17], [20], although further reduction can occur with additional exposures [21], [22].

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B. Theories of CARE

Several theoretical accounts of cybersickness and CARE have been proposed in the literature. These accounts provide useful context for understanding why CARE may or may not generalize across VR experiences. Although the present study was not designed to adjudicate between these accounts, summarizing the major perspectives helps frame the types of generalization patterns that might arise under different assumptions.

Sensory conflict theory [23] proposes that sickness can occur when multiple sensory inputs conflict with each other or with past experience. For example, a seated VR user might experience visual input indicating forward self-motion while vestibular and proprioceptive input indicates lack of motion. These conflicting inputs and their mismatch with prior real-world locomotor experience are thought to produce cybersickness.

Sensory rearrangement theory [24] builds on sensory conflict theory by proposing that novel sensory arrangements can be learned and incorporated into an internal model that predicts sensory consequences of actions. According to this view, CARE reflects refinement of an internal model that links user actions (e.g., joystick input, arm movements) to resulting sensory outcomes. When the locomotion interface changes, the internal model may no longer match the new sensory outcomes, causing sickness to reemerge until the model is further updated or a new model is constructed.

The internal model was originally described as the ”neural store” [24] and later framed as an ”internal dynamic model” [25]. A person may learn multiple internal models for different contexts and deploy them appropriately depending on environmental cues, task characteristics, and locomotion interface. Because locomotion interfaces exert a strong influence on visually induced motion and sickness [26], [27], they may also strongly constrain whether an internal model learned in one VR context will generalize to another.

Habituation has also been proposed as an explanation for CARE [28]. Habituation refers to a reduced response to repeated exposure to a stimulus and may occur in response to persistent sickness-inducing elements of VR such as latency, sensory conflict, or optic flow. Unlike sensory adaptation, which reflects changes in peripheral sensory neuron sensitivity, habituation reflects central nervous system learning and unfolds over repeated exposures in a manner consistent with observed patterns of CARE. Because habituation is thought to operate on sickness-inducing stimulus properties (e.g., latency, optic flow) that can be shared across experiences, it naturally predicts some degree of CARE generalization when those properties are similar.

In the present study, these theoretical accounts are used to frame possible patterns of generalization, rather than as directly tested hypotheses.

C. Generalization of CARE

Ideally, CARE that occurs in one VR context will generalize broadly to other VR contexts (i.e., different environments, different input devices, etc.). Such broad generalization of CARE would prevent the need to experience initial sickness before CARE occurs within every new VR context. Yet, evidence of generalization is mixed.

In one study [16], participants experienced a passive VR rollercoaster simulation twice on separate days. Participants also experienced an active VR climbing game, half of them after the first rollercoaster exposure (coaster-climb-coaster) and half after the second rollercoaster exposure (coaster-coaster-climb). In both conditions, cybersickness ratings after the second rollercoaster exposure were about 35% lower compared to the first rollercoaster exposure, which is evidence of CARE. However, cybersickness caused by the second rollercoaster exposure was no lower for the group that additionally experienced the climbing game after the first rollercoaster exposure (coaster-climb-coaster) compared to the group that did not experience the climbing game until after the rollercoaster (coaster-coaster-climb). In other words, any CARE that might have occurred during the climbing game in the coaster-climb-coaster group did not generalize to the second rollercoaster exposure. Furthermore, cybersickness experienced during the climbing game did not differ between groups, indicating that additional CARE that may have occurred during the second rollercoaster exposure in the coaster-coaster-climb group did not generalize to the climbing game.

In another study [14], participants experienced the same VR game on four separate days. On Days 1-3 they experienced the game with field-of-view (FOV) restriction, which reduces cybersickness by blocking peripheral visual motion [20], [29], [30], [31], [32]. Cybersickness was lower on Day 3 compared to Day 1, which is evidence of CARE when using FOV restriction. When participants played the same game with unrestricted FOV on Day 4, cybersickness remained low and was significantly lower than a control group that experienced unrestricted FOV on a single day. In other words, CARE that occurred with FOV restriction generalized to a situation without FOV restriction.

The two studies reviewed in this section reach distinct conclusions about generalization of CARE, yet the studies employ different methodologies that likely underlie the distinct conclusions. For example, one study [16] investigated generalization of CARE across many contextual changes, including the virtual environment, locomotion interface, and user task, whereas the other study [14] investigated generalization of CARE across just one contextual change, FOV. These and other methodological differences may be responsible for the distinct conclusions about generalization of CARE.

Although CARE itself occurs within an individual, research on generalization of CARE requires comparing groups with different exposure histories. A purely within-participant approach (e.g., a pre-test/post-test design in which participants experience one VR simulation before and after repeated exposure to a different one) cannot isolate generalization. Any reduction in sickness during the second exposure could be caused by simply repeating the same simulation (i.e., from pre-test to post-test), making it impossible to determine whether the change reflects generalization or within-simulation CARE. For this reason, prior studies of generalization have relied on between-participant comparisons in which one group has prior VR exposure and a comparison group does not.

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D. Study Overview and Hypotheses

The purpose of the current study was to explore the practical question of whether CARE generalizes across VR experiences with distinct contexts, including different tasks, environments, and locomotion interfaces.

Testing generalization requires two groups with different exposure histories: one group (Repeated Exposure) that encounters the target simulation after previously experiencing repeated exposure to another VR context, and a comparison group (Single Exposure) that encounters the target simulation for the first time, thereby avoiding contamination from repeating the target simulation. Accordingly, participants in the Repeated Exposure conditions were exposed to VR on four separate days. On Days 1-3, they played either a puzzle game (Shadow Point) that used joystick-based locomotion or a climbing game (Climb 2) that used arm movements to negotiate handholds. On Day 4, Repeated Exposure participants played a stealth survival game (Jurassic World) that used joystick input. Participants in the Single Exposure condition played the survival game only once.

The puzzle game and survival game used a very similar joystick locomotion interface. In contrast, the climbing game and survival game used different locomotion interfaces. All three games differed in environment, task, and game mechanics. These similarities and differences afforded the opportunity to examine whether CARE generalizes across changes in VR context and, in particular, whether it generalizes across changes in the locomotion interface, which may play an especially important role in CARE (see Section II-B).

Theoretical accounts of CARE make distinct predictions for the two Repeated Exposure conditions. If CARE occurs because users learn and deploy an internal model of a locomotion interface (sensory rearrangement theory), then participants who experience a similar locomotion interface on Day 4 should show greater generalization of CARE compared to those who experience a different interface. If CARE reflects habituation to VR characteristics shared across experiences, then both Repeated Exposure conditions should show similar generalization on Day 4, regardless of interface similarity. These accounts are not mutually exclusive; it is possible that general habituation produces broad CARE while interface similarity confers additional benefits. These conditions therefore allow us to examine whether generalization patterns align more closely with predictions from sensory rearrangement or habituation accounts (Section II-B), while recognizing that the study was not designed to serve as a direct test of either theory.

III. Method

A. Participants

Two hundred and twenty-eight undergraduate students (115 men, 113 women) at Iowa State University participated in exchange for course credit. Of these, 114 participants were assigned to the Single Exposure condition, 62 were assigned to the Repeated Exposure: Shadow Point condition, and 52 were assigned to the Repeated Exposure: Climb 2 condition (herein referred to as Repeated Exposure: Climb for ease of exposition).

TABLE I
Participant Demographics by Condition (SP = Shadow Point) and Gender

Condition	Gender	Count	Game hrs/wk	VIMSSQ	MSSQ
Repeated: Climb	Male	17	36.26 (38.18)	4.22 (3.05)	34.67 (31.43)
	Female	19	10.78 (12.00)	7.00 (4.85)	78.68 (67.52)
	Total	36	22.81 (30.11)	5.68 (4.28)	57.90 (57.38)
Repeated: SP	Male	16	21.18 (12.79)	2.37 (2.02)	16.75 (19.32)
	Female	21	14.66 (15.15)	4.38 (3.04)	30.58 (26.07)
	Total	37	17.48 (14.37)	3.51 (2.80)	24.6 (24.11)
Single Exposure	Male	59	28.49 (28.89)	3.10 (2.65)	29.63 (38.17)
	Female	55	11.12 (15.15)	4.09 (3.04)	38.86 (26.07)
	Total	114	20.11 (21.27)	3.57 (2.95)	34.08 (38.15)

The sample size for Single Exposure participants was higher because data collected in that condition were also relevant to other ongoing projects that require a larger sample size. Due to participant dropout in the Repeated Exposure condition, 37 participants completed the Repeated Exposure Shadow Point condition and 36 participants completed the Repeated Exposure Climb condition. Incomplete participants in the Repeated Exposure conditions were asked to report their reasons for withdrawal and they primarily cited schedule conflicts. Data from incomplete participants were excluded from analysis (but see Section IV-B, which compares demographic and cybersickness data of participants who did and did not drop out from the Repeated Exposure conditions to evaluate potential bias).

Demographic information is displayed in Table I. The age range of the remaining 187 participants was 18 to 28 (M = 19.5, SD = 1.8) and consisted of 92 males (49.2%). Although relatively few participants had previous VR experience (11.8%), most had experience with video games (94.7%; M = 20.1 gaming hours per week, SD = 21.5). The short form Motion Sickness Susceptibility Questionnaire (MSSQ) [33] indicated that this sample's susceptibility to motion sickness was typical (M = 36.4, SD = 41.3). The short form Visually Induced Motion Sickness Questionnaire (VIMSSQ-short) [11], [34] lacks normative data with which to compare data from current sample (M = 3.9, SD = 3.3).

B. Stimuli and Design

The experiment utilized three VR games purchased from the Meta Quest store and presented via Meta Quest head-mounted displays (resolution 1440 × 1600 pixels per eye, 72 Hz refresh rate, and approximately 95° field of view). Jurassic World: Aftermath is a stealth survival game in which the player attempts to escape an abandoned research island while evading dinosaurs. Locomotion is accomplished via joystick-based translation and rotation. Shadow Point is a puzzle game in which the player manipulates objects and lighting to create patterns in cast shadows, using joystick-based translation and rotation to navigate the environment. The Climb 2 is a climbing game in which the player traverses obstacles such as cliffs and bridges by grasping a sequence of hand-holds, with locomotion driven by arm movements while holding the controllers. All three games were rated "moderate" comfort on the Quest store, a platform-assigned,

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qualitative classification for user comfort rather than a standardized or quantitative measure of sickness potential. Informal pilot testing in the lab indicated that all three games elicited noticeable but not extreme cybersickness symptoms among lab members.

Although the games were not instrumented to quantify visual-motion parameters, they differed qualitatively in their dominant optic flow and rotation characteristics. Jurassic World: Aftermath involved frequent low-to-moderate forward translation through enclosed corridors, with intermittent joystick-based yaw rotations during navigation and evasion. Shadow Point emphasized slower, more deliberate translation in comparatively open environments, with less frequent rotations as participants repositioned to manipulate objects and shadows. In contrast, The Climb 2 primarily elicited visually induced motion through vertical and lateral viewpoint shifts driven by arm movements, with minimal continuous optic flow but frequent changes in viewpoint orientation as participants reached between holds.

There were three experimental conditions, two of which involved repeated exposure to VR and one that involved a single exposure. In the Single Exposure condition, participants played Jurassic World on one day. In the Repeated Exposure conditions, participants played Climb 2 or Shadow Point on Days 1-3 and played Jurassic World on Day 4.

C. Measures of Cybersickness

Cybersickness was measured in multiple ways. First, participants completed the Simulator Sickness Questionnaire (SSQ) [2], a 16-item survey of potential cybersickness symptoms, each rated on a 4-point scale from 0 (none) to 3 (severe). The SSQ was administered before and after VR exposure, and responses were used to calculate subscales and a total score following the original recommendations [2].

Second, participants rated their sickness every 4 minutes during VR exposure and again upon exiting VR using an 11-point scale from 0 (none) to 10 (severe), a modified version of the 21-point fast motion sickness (FMS) scale [35]. The scale range was modified from the original range for ease of comprehension. Despite this change, this measure is herein referred to as FMS. The measure is similar to the 11-point discomfort score used in cybersickness research [20], [30], [36]. The primary data point for statistical analysis was the final FMS measurement taken at the end of VR exposure.

Third, exposure time was measured from the start of VR exposure to the end of VR exposure (either when 20 minutes elapsed or when the participant withdrew).

FMS, SSQ total, and exposure time were the primary indicators of cybersickness; additional analyses of SSQ subscales were treated as exploratory and are provided on the Open Science Framework [37].

D. Procedure

Upon arrival at the lab on their first day (i.e., Single Exposure condition and Day 1 of Repeated Exposure conditions), the participant reviewed and signed the informed consent document. The participant then completed several pre-exposure survey measures: demographics, a video game usage questionnaire, the short-format MSSQ [38], the short-format VIMSSQ [11], and the SSQ [2]. The participant was then given basic information about the game they would be playing and how to use the controllers (e.g., which controller movements corresponded to which actions in the game). This training phase was intentionally brief (approximately 2-3 minutes) and conducted outside of the VR headset to avoid the potential for cybersickness symptoms occurring prior to the experimental exposure. The researcher then measured the participant's interpupillary distance (IPD) with a ruler, adjusted the headset IPD to the value nearest the participant's measured IPD, and then helped the participant don and adjust the fit of the headset.

The participant was instructed to play the game for up to 20 minutes or until they could no longer continue due to cybersickness. The participant's view in the headset was cast to a nearby monitor, allowing the researcher to monitor progress and to provide assistance with challenges (e.g., reminders of controller movements or guidance through challenging sections of the game) as needed. The researcher asked the participant to rate their sickness on a scale from 0 (none) to 10 (severe) every 4 minutes during VR exposure, starting upon initial entry into VR. A final measurement was taken when the participant exited VR. After exiting VR, the participant completed the SSQ and the virtual reality neuroscience questionnaire (VRNQ) [39], which included several 7-point ratings of enjoyment and usability (cybersickness questions on the VRNQ were excluded due to redundancy with SSQ items).

For those in the Repeated Exposure conditions, subsequent days differed from the first day in the following ways: Pre-exposure surveys (other than the SSQ) were not administered beyond the first day, and the VRNQ was not administered again until Day 4 following exposure to Jurassic World.

E. Statistical Analysis

Analyses were conducted to address two primary questions: (1) whether cybersickness abated across repeated exposures within the Repeated Exposure conditions (CARE), and (2) whether any abatement generalized to a novel VR experience relative to a Single Exposure control condition.

To evaluate CARE within each Repeated Exposure condition (i.e., The Climb 2 and Shadow Point), repeated-measures ANOVAs were conducted with exposure day (Days 1-3) as a within-participants factor. Separate models were fit for each game and for each primary dependent variable: final FMS rating, exposure time, and SSQ total score. When Mauchly's test indicated violations of sphericity, Greenhouse-Geisser corrections were applied.

To evaluate generalization of CARE to a novel VR experience, one-way ANCOVAs were conducted on cybersickness outcomes measured during exposure to Jurassic World, with Condition (Repeated Exposure: Climb, Repeated Exposure: Shadow Point, Single Exposure) as the between-participants factor. Gender, prior VR experience, VIMSSQ score, MSSQ score, and self-reported gaming hours per week were included as covariates. These covariates were selected based on prior literature and observed correlations with cybersickness outcomes in the present

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TABLE II
CORRELATIONS, MEANS, AND STANDARD DEVIATIONS FOR KEY DEMOGRAPHIC MEASURES AND CYBERSICKNESS MEASURES. CYBERSICKNESS DATA ARE FROM EXPOSURE TO JURASSIC WORLD, WHICH OCCURRED ON DAY 4 FOR REPEATED EXPOSURE PARTICIPANTS.

	1.	2.	3.	4.	5.	6.	7.	8.	9.	M	SD
1. Gender	—									—	—
2. IPD	.45**	—								58.88	4.71
3. VR	.11	.03	—							12.00	32.30
4. Game hrs / wk	.38**	.17*	.05	—						20.12	22.10
5. VIMSSQ	−.23**	−.05	−.20**	−.21**	—					3.97	3.32
6. MSSQ	−.20**	.01	−.13	−.18*	.59**	—				36.80	41.68
7. FMS	−.31**	−.09	−.19**	−.27**	.40**	.27**	—			5.40	3.07
8. Time (min)	.12	−.01	.15*	.13	−.40**	−.29**	−.61**	—		15.31	5.29
9. SSQ Total	−.12	.03	−.16*	−.10	.41**	.32**	.66**	−.52**	—	55.86	42.94

TABLE III
MEANS (AND STANDARD ERRORS) OF CYBERSICKNESS OUTCOMES AFTER PLAYING JURASSIC WORLD FOR EACH CONDITION. VALUES ARE COVARIATE-ADJUSTED FOR CONSISTENCY WITH THE REPORTED ANALYSES.

Condition	FMS	Time played (min)	SSQ Total
Repeated: Climb	4.56 (0.44)	15.81 (0.78)	51.37 (6.40)
Repeated: SP	4.85 (0.45)	16.85 (0.78)	43.14 (6.43)
Single Exposure	5.88 (0.25)	14.83 (0.44)	61.45 (3.65)

sample (Table II), and were included to control for individual differences known to influence cybersickness.

For dependent variables that exhibited heterogeneity of variance or departures from normality, rank-based ANCOVAs and nonparametric sensitivity analyses were conducted to assess the robustness of the primary findings. In addition, to address unequal group sizes and potential imbalance in individual characteristics, a covariate-adjusted, gender-stratified bootstrap procedure was used to compare each Repeated Exposure group to matched subsamples drawn from the Single Exposure group. Bootstrap analyses were conducted with 10,000 resamples.

Throughout the manuscript, exact p-values are reported to three decimal places, except when $p < 001$, in which case results are reported as $p < 001$.

IV. RESULTS

Results are presented first regarding the question of whether CARE occurred during the repeated conditions over Days 1-3 (Section IV-C). Then results are presented regarding CARE generalization on Day 4 (Section IV-D).

Raw data are available on the Open Science Framework [37]. Demographics (gender, gaming hours per week, MSSQ, and VIMSSQ) are shown in Table I, and cybersickness measures (final FMS, exposure time in minutes, and SSQ total) are shown in Table III.

The means, standard deviations, and correlations among individual difference measures and key measures of cybersickness are shown in Table II. The relatively large standard deviations observed in Table I, particularly for gaming hours and MSSQ, reflect substantial heterogeneity in individual experience and motion sickness susceptibility within this sample. Subsequent statistical analyses that compare across conditions include gender, gaming experience, VR experience, VIMSSQ, and MSSQ as covariates. A chi-square test of independence indicated that gender distribution did not significantly differ across conditions, $\chi^2(2) = 0.88, p = .640$.

Given prior evidence that women are at greater risk for cybersickness [8], [9], and the significant correlations between gender and FMS observed in our sample, we conducted additional exploratory analyses treating gender as a between-participants factor. For the CARE analyses (Days 1-3 of the Repeated Exposure conditions), we conducted mixed ANOVAs with day as a within-participants factor and gender as a between-participants factor for each game. For the generalization analyses (Jurassic World only; Single Exposure condition and Day 4 of the Repeated Exposure conditions), we extended the ANCOVAs to include a condition by gender term in addition to the existing covariates.

Across analyses, gender frequently showed the expected main effect on cybersickness, with women experiencing greater sickness than men. For the CARE assessment (Days 1-3), the only reliable moderation effect was a gender by day interaction for exposure time in the Climb 2 condition, reflecting relatively shorter play times for women on later days. For the generalization phase, a modest condition by gender interaction emerged for FMS: among participants who had previously played Shadow Point, men reported lower cybersickness when later playing Jurassic World compared to women, whereas the gender difference was smaller in the Climb and Single Exposure groups. No gender interactions were observed for SSQ total or exposure time. Because these moderation effects were exploratory, inconsistent across outcomes, and not central to our primary hypotheses, full details are provided on the Open Science Framework [37].

A. Assumption Checks and Robustness Analyses

Before reporting the results of analyses evaluating changes in sickness across repeated exposures (Section IV-C) and the generalization of CARE to a novel VR experience (Section IV-D), a series of checks were conducted to evaluate whether the assumptions of the primary statistical tests were adequately met for the primary dependent variables (FMS, SSQ total, and exposure time). Detailed results of these checks are available on the Open Science Framework [37].

1) Repeated-Measures ANOVAs (CARE Analyses): For the CARE analyses in Section IV-C, repeated-measures ANOVAs

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were conducted with exposure day (Days 1-3) as a within-participants factor, separately for each Repeated Exposure condition (Climb 2 and Shadow Point). Residual normality was assessed using Shapiro–Wilk tests. Where deviations from normality were observed, they were mild and did not materially affect model fit.

Sphericity was evaluated using Mauchly's test. When this assumption was violated, Greenhouse–Geisser corrections were applied and are reported with the corresponding test statistics.

To assess robustness to potential assumption violations, nonparametric Friedman tests were conducted for each dependent variable and condition. Across all measures, the pattern of significance matched the parametric ANOVA results, supporting the robustness of the reported findings.

2) ANCOVAs (Generalization Analyses): For the generalization analyses in Section IV-D, assumptions of normality, homogeneity of variances, and linearity of covariate effects were evaluated for each ANCOVA model. Normality of residuals was assessed using Shapiro–Wilk tests and visual inspection of residual distributions. Levene's tests were used to assess homogeneity of variances across conditions.

For dependent variables that showed heterogeneity of variance or departures from normality, rank-based ANCOVAs were conducted as sensitivity analyses. These analyses reproduced the same pattern of results as the parametric models.

Linearity and homogeneity of regression slopes were evaluated using covariate-by-condition interaction terms and visual inspection of scatterplots, confirming that covariate effects did not differ significantly across conditions.

B. Assessment of Attrition Bias

To determine if the 41 participants who did not complete all four days in the Repeated Exposure conditions differed from the 73 participants who did complete all four days, an independent samples t-test was conducted on key demographics, final FMS, exposure time, and SSQ total. These comparisons used only Day 1 data. No significant differences in demographics existed between those who completed all four days and those who did not. Additionally, there were no significant differences in Day 1 cybersickness levels between those who completed four days and those who did not. These analyses are available on the Open Science Framework [37].

C. Cybersickness Abatement From Repeated Exposure (CARE)

To evaluate CARE in the Repeated Exposure conditions, cybersickness data from Days 1-3 were analyzed in repeated measures ANOVAs. Analyses were conducted separately for the two Repeated Exposure conditions (Climb 2 and Shadow Point). Assumption checks and a nonparametric Friedman-test sensitivity analysis are reported in Section IV-A. Distributional summaries for these repeated-exposure measures are provided in the supplementary materials on the Open Science Framework [37].

1) FMS: For Climb 2, final FMS ratings (see Fig. 1, dashed line) showed a significant effect of day, $F(2, 70) = 5.36$, $p = .007$, $\eta_p^2 = .133$. Contrary to expectations, FMS ratings were significantly higher on Day 2 compared to Day 1, $F(1, 35) = 6.95$, $p = .012$, $\eta_p^2 = .166$, whereas Day 3 did not significantly differ from Day 2, $F(1, 35) = 0.06$, $p = .805$, $\eta_p^2 = .002$. FMS ratings on Day 3 were also significantly higher than Day 1, $F(1, 35) = 7.36$, $p = .010$, $\eta_p^2 = .174$.

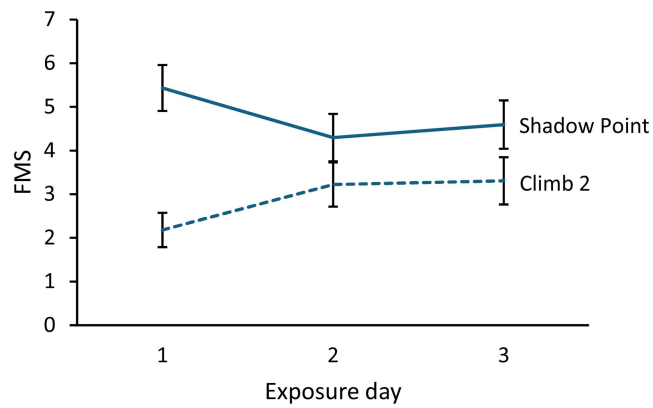

Fig. 1. Mean final FMS by exposure day and game for participants in the Repeated Exposure conditions. Error bars represent ±1 SEM. Supplemental distributional summaries for these measures are provided on OSF [37].

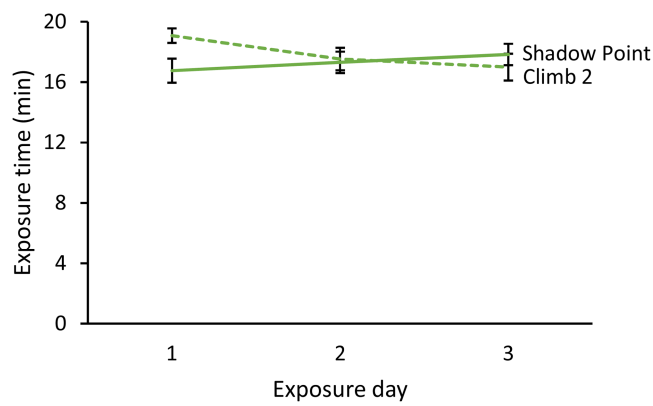

Fig. 2. Mean exposure time by exposure day and game for participants in the Repeated Exposure conditions. Error bars represent ±1 SEM. Supplemental distributional summaries for these measures are provided on OSF [37].

For the Repeated Exposure: Shadow Point condition (see Fig. 1, solid line), the main effect of day was significant, $F(2, 72) = 5.48$, $p = .006$, $\eta_p^2 = .132$. FMS ratings were significantly lower on Day 2 compared to Day 1, $F(1, 36) = 9.41$, $p = .004$, $\eta_p^2 = .207$, whereas Day 3 did not significantly differ from Day 2, $F(1, 36) = 0.66$, $p = .421$, $\eta_p^2 = .018$. FMS ratings on Day 3 were also significantly lower than Day 1, $F(1, 36) = 6.45$, $p = .016$, $\eta_p^2 = .152$.

2) Exposure Time: For Climb 2, exposure time (see Fig. 2, dashed line) showed a significant effect of day, $F(2, 70) = 5.74$, $p = .005$, $\eta_p^2 = .141$. Contrary to expectations, exposure time was significantly lower on Day 2 compared to Day 1, $F(1, 35) = 9.30$, $p = .004$, $\eta_p^2 = .210$, whereas Day 3 did not significantly differ from Day 2, $F(1, 35) = 0.57$, $p = .454$, $\eta_p^2 = .016$. Exposure time on Day 3 was also significantly lower than Day 1, $F(1, 35) = 9.23$, $p = .004$, $\eta_p^2 = .209$.

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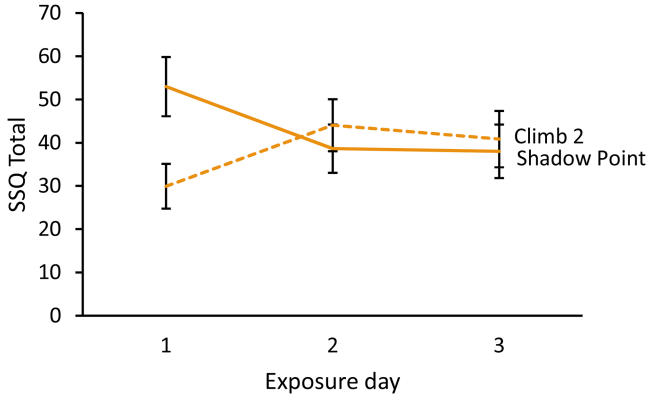

Fig. 3. Mean SSQ total scores by exposure day and game for participants in the Repeated Exposure conditions. Error bars represent ±1 SEM. Supplemental distributional summaries for these measures are provided on OSF [37].

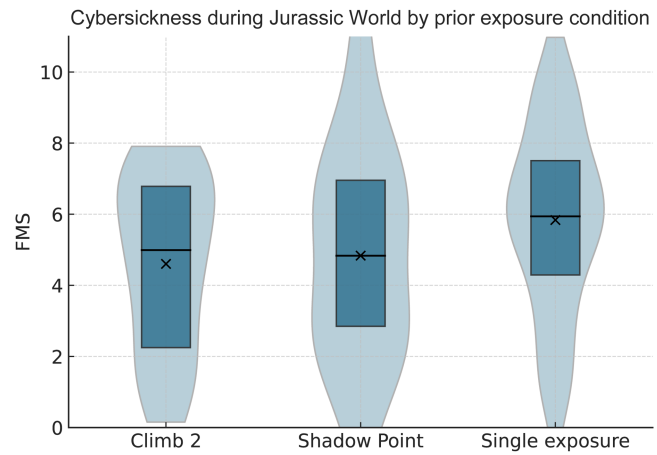

Fig. 4. Violin and box plots showing final FMS (covariate-adjusted) when playing Jurassic World for each condition. Violin plots show the full distribution density; box plots indicate the interquartile range (25th–75th percentiles) with the horizontal line marking the median; whiskers extend to 1.5 × IQR; black × symbols indicate group means.

For the Repeated Exposure: Shadow Point condition (see Fig. 2, solid line), Mauchly's test of sphericity was violated, so the analysis proceeded using a Greenhouse-Geisser correction. The main effect of day was not significant, $F(1.53, 54.95) = 1.75$, $p = .181$, $\eta_p^2 = .046$. Exposure time did not significantly differ between Day 1 and Day 2, $F(1, 36) = 0.68$, $p = .416$, $\eta_p^2 = .018$, nor between Day 2 and Day 3, $F(1, 36) = 0.70$, $p = .408$, $\eta_p^2 = .019$. Exposure time on Day 3 was significantly greater than Day 1, $F(1, 36) = 7.70$, $p = .009$, $\eta_p^2 = .176$.

3) SSQ: The SSQ was administered before and after exposure to VR. SSQ sub-scores (nausea, oculomotor, and disorientation) and total score were calculated using the recommendations from the original paper [2]. Pre-exposure scores indicated no lingering symptoms from prior exposure on Days 2-4 of the Repeated Exposure conditions, and these analyses are available on the Open Science Framework [37]. All subsequent analyses focus on post-exposure scores, which were used to evaluate cybersickness caused by VR exposure. In the main manuscript, we report results for the SSQ total score, shown in Fig. 3. Detailed analyses of each SSQ subscale are available on the Open Science Framework [37].

For SSQ total data in the Repeated Exposure: Climb condition (see Fig. 3, dashed line), Mauchly's test of sphericity was violated, so the analysis proceeded using a Greenhouse-Geisser correction. The main effect of day was significant, $F(1.57, 54.87) = 5.46$, $p = .011$, $\eta_p^2 = .135$. SSQ total was significantly higher on Day 2 compared to Day 1, $F(1, 35) = 9.55$, $p = .004$, $\eta_p^2 = .214$, whereas Day 3 did not significantly differ from Day 2, $F(1, 35) = 0.98$, $p = .329$, $\eta_p^2 = .027$. SSQ total on Day 3 was also significantly higher than Day 1, $F(1, 35) = 4.13$, $p = .050$, $\eta_p^2 = .106$.

For the Repeated Exposure: Shadow Point condition (see Fig. 3, solid line), Mauchly's test of sphericity was violated so the analysis proceeded using a Greenhouse-Geisser correction. The main effect of day was significant, $F(1.71, 61.60) = 9.72$, $p < 001$, $\eta_p^2 = .213$. SSQ total was significantly lower on Day 2 compared to Day 1, $F(1, 36) = 10.49$, $p = .003$, $\eta_p^2 = .226$, whereas Day 3 did not significantly differ from Day 2, $F(1, 36) = 0.04$, $p = .843$, $\eta_p^2 = .001$. SSQ total on Day 3 was also significantly lower than Day 1, $F(1, 36) = 14.54$, $p < 001$, $\eta_p^2 = .288$.

D. Generalization of CARE

For each dependent variable (FMS, exposure time, SSQ total), we conducted a one-way ANCOVA with Condition (Climb 2, Shadow Point, Single Exposure) and covariates gender, VR experience, VIMSSQ, and MSSQ. All cybersickness data considered in these analyses were based on the same game (Jurassic World: Aftermath), and any differences can be attributed to the prior VR exposure that occurred in the Repeated Exposure conditions. Assumption checks for these ANCOVAs are provided in Section IV-A. Estimated marginal means for each of the conditions and variables are displayed in Table III. In exploratory follow-up analyses, gender was also treated as a between-participants factor (rather than solely as a covariate) to test for possible condition by gender interactions.

To address concerns about unequal group sizes and individual differences, a covariate-adjusted, gender-stratified bootstrap analysis was conducted to compare each Repeated Exposure group to equally sized, covariate-matched subsamples drawn from the Single Exposure pool. Within the Single Exposure group, outcomes were first adjusted for the ANCOVA covariates (gender, VIMSSQ, MSSQ); residuals were then resampled 10,000 times using gender-stratified sampling to match each repeated-exposure group in sample size and gender composition. The bootstrap results supported the main ANCOVA findings and are reported on the Open Science Framework [37].

1) FMS: The homogeneity of variances assumption was met, $F(2, 184) = 1.85$, $p = .160$. Estimated marginal means for each of the conditions are displayed in Fig. 4. The main effect of condition was statistically significant, $F(2, 179) = 3.53$, $p = .031$, $\eta_p^2 = .038$. Repeated contrasts indicated that FMS in the Single Exposure condition was significantly higher than that in the Repeated Exposure: Climb condition ($p = .036$) as well as

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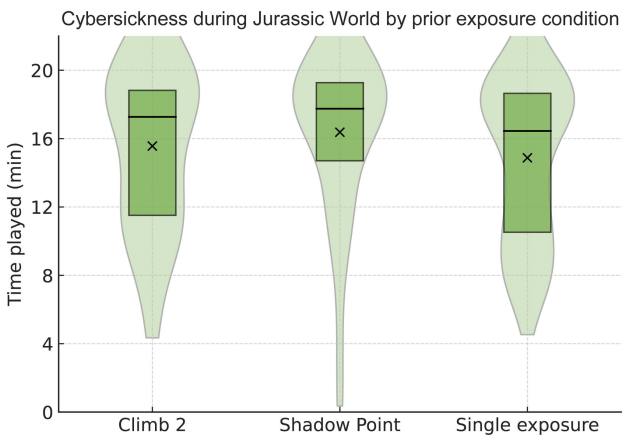

Fig. 5. Violin and box plots showing exposure time (covariate-adjusted) when playing Jurassic World for each condition. Violin plots show the full distribution density; box plots indicate the interquartile range (25th–75th percentiles) with the horizontal line marking the median; whiskers extend to 1.5 × IQR; black × symbols indicate group means.

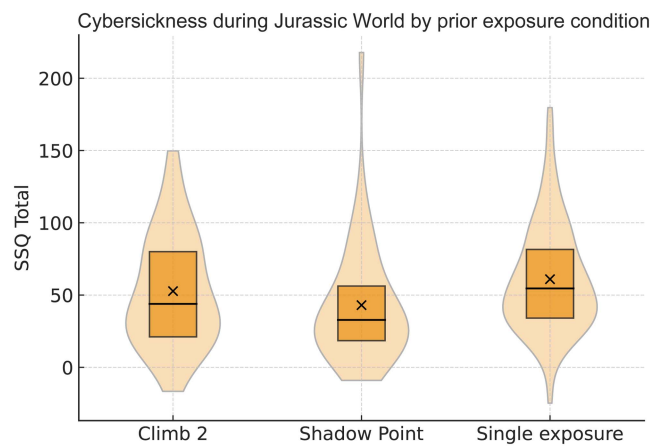

Fig. 6. Violin and box plots showing SSQ total scores (covariate-adjusted) when playing Jurassic World for each condition. Violin plots show the full distribution density; box plots indicate the interquartile range (25th–75th percentiles) with the horizontal line marking the median; whiskers extend to 1.5 × IQR; black × symbols indicate group means.

the Repeated Exposure: Shadow Point condition ($p = .045$). The Repeated Exposure conditions did not significantly differ from one another ($p = .880$). Covariates VIMSSQ, $F(1, 179) = 15.65$, $p < 001$, $\eta_p^2 = .080$, and gender, $F(1, 179) = 48.90$, $p = .010$, $\eta_p^2 = .037$, were significant. No other covariates were significant.

2) Exposure Time: The homogeneity of variances assumption was met, $F(2, 184) = 1.03$, $p = .360$. Estimated marginal means for each of the conditions are displayed in Fig. 5. The main effect of condition was not significant, $F(2, 179) = 1.38$, $p = .255$, $\eta_p^2 = .015$. Analysis proceeded with repeated contrasts to test specific hypotheses despite the lack of a significant main effect. Exposure time in the Single Exposure condition did not differ significantly from that in the Repeated Exposure: Climb condition ($p = .524$) nor the Repeated Exposure: Shadow Point condition ($p = .104$). The Repeated Exposure conditions did not significantly differ from one another ($p = .453$). Covariate VIMSSQ was significant, $F(1, 179) = 15.78$, $p < 001$, $\eta_p^2 = .081$. No other covariates were significant.

3) SSQ: The homogeneity of variances assumption was met, $F(2, 184) = 0.19$, $p = .827$. Estimated marginal means for each of the conditions are displayed in Fig. 6. The main effect of condition was statistically significant, $F(2, 179) = 3.09$, $p = .048$, $\eta_p^2 = .033$. Repeated contrasts indicated that SSQ total in the Single Exposure condition was significantly higher than that in the Repeated Exposure: Shadow Point condition ($p = .017$) but did not significantly differ from that in the Repeated Exposure: Climb condition ($p = .290$). The Repeated Exposure conditions did not significantly differ from one another ($p = .309$). Covariate VIMSSQ was significant, $F(1, 179) = 15.39$, $p < 001$, $\eta_p^2 = .079$. No other covariates were significant.

V. Discussion

The primary goal of this study was to explore whether prior VR exposure generalizes to reduce cybersickness in another VR experience with a distinct task, environment, and locomotion interface. Generalization across VR contexts would mean that the reduced sickness achieved through exposure to one VR context could transfer to another VR context, thereby reducing the initial discomfort that the user would otherwise have experienced in the new context. A secondary goal was to interpret patterns of generalization in light of theoretical accounts that have been proposed to explain why CARE occurs. These theoretical perspectives, such as sensory rearrangement and habituation, remain active in the literature. Although the present study was not designed to distinguish between theories, they provide useful context for interpreting how and why generalization might occur across distinct VR stimuli.

Cybersickness ratings on the FMS scale were lower on Day 4 compared to the Single Exposure condition, and this was true for both Repeated Exposure games (Shadow Point and Climb 2). This finding indicates that CARE that occurred during exposure to the puzzle game and the climbing game generalized to the survival game, leading to lower cybersickness. The practical implication of this finding is that CARE that occurs in one VR context can generalize to a novel context with a distinct environment, task, and locomotion interface. The Climb 2 involved traversing virtual obstacles by moving one's arms to grasp handholds. In comparison, Jurassic World represents a completely different experience in which the user manipulates a joystick to navigate through an abandoned lab facility. Shadow Point used a locomotion interface similar to that used in Jurassic World, but the task and environment differed considerably.

Cybersickness ratings on the SSQ scale (specifically the SSQ total score and the SSQ nausea subscale) were also lower on Day 4 compared to the Single Exposure condition, but only in the Repeated Exposure: Shadow Point condition, and not in the Repeated Exposure: Climb condition. The divergence between FMS and SSQ outcomes may reflect differences in the constructs captured by the two measures. To further examine this discrepancy, we conducted exploratory ANCOVAs on SSQ symptom subscales (nausea, oculomotor, and disorientation) for the Day 4 exposure (Jurassic World), using the same covariates as in the primary analyses. No reliable main effect of condition

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emerged for any individual subscale (all $p > .10$), indicating that generalization effects observed in SSQ total were not driven by a single symptom domain. Instead, these results suggest that SSQ total effects reflect modest, distributed changes across multiple symptom categories, whereas the unidimensional FMS appears more sensitive to global changes in sickness severity. Full subscale analyses are available on the Open Science Framework [37].

Taken together, the FMS and SSQ results suggest that CARE generalization in the present study is most consistently expressed at the level of global, subjective sickness severity rather than within any single symptom domain. The robust Day 4 effects observed for FMS across both Repeated Exposure conditions, combined with weaker and less consistent effects for SSQ total and null effects at the subscale level, indicate that prior VR exposure primarily shifts overall tolerance, rather than producing large, symptom-specific reductions. This pattern provides a coherent account of why generalization was clearly detected with a unidimensional global measure (FMS), while appearing more fragile when assessed with a multi-symptom measure (SSQ).

Although CARE was evident in Shadow Point, it was weak or absent in Climb, where sickness increased from Day 1 to Day 2 on some measures. These mixed patterns may reflect uncontrolled variation in how task demands and visual-motion exposure evolve as participants progress through the commercial games. Although game progression and difficulty were not formally measured in the present study, Shadow Point typically introduces new mechanics and puzzles gradually, which may stabilize or reduce visual-motion demands across sessions. In contrast, Climb can expose participants to increasingly complex routes and greater vertical displacement (including falls) as progression occurs, potentially increasing optic flow and task difficulty across days. Such changes could counteract adaptation effects. Importantly, this inconsistency does not compromise the primary test of CARE generalization: because Day 4 outcomes were measured on the same game (Jurassic World) across all conditions, differences on Day 4 can be attributed to exposure history even when within-game CARE trajectories vary. The observation that generalization occurred despite weak within-game CARE in one of the repeated-exposure games suggests that early VR experience may benefit users even when symptom abatement is modest.

Despite the uncontrolled stimulus intensity during repeated exposure, the use of off-the-shelf consumer games adds considerable ecological validity to cybersickness research. The games used in the current study are owned by thousands of VR users,[1] and the results therefore reflect experiences that may actually occur among the broader population of VR users. The use of consumer games also means that visual-motion cues were not experimentally controlled. As a result, the present findings do not isolate which specific components of visual motion, such as linear optic flow or angular rotation, support or limit CARE or its generalization. Controlled, parametric motion studies (e.g., [40]) will be essential for identifying the motion cues that drive these effects and for determining how generalization manifests under systematically varied visual-motion profiles.

Theoretical accounts of CARE make distinct predictions about generalization of CARE in the two Repeated Exposure conditions. Sensory rearrangement theory [24], an extension of sensory conflict theory [23], proposes that novel sensory arrangements (e.g., the sensory consequences of controller inputs within a virtual environment) are learned and form an internal model that predicts sensory outcomes of actions. According to this theory, a change to the locomotion interface on Day 4 should cause cybersickness to reemerge because the new locomotion interface imposes a novel sensory arrangement that must be learned. Shadow Point and Jurassic World use very similar joystick-based locomotion interfaces, such that the internal model of joystick locomotion learned when playing Shadow Point should be useful for predicting the sensory outcomes of joystick inputs in Jurassic World. In contrast, Climb 2 uses a distinct climbing interface with completely different inputs (e.g., arm movements rather than joystick manipulation). Therefore, sensory rearrangement theory would predict Day 4 sickness when playing Jurassic World should be higher after prior exposure to Climb 2 compared to prior exposure to Shadow Point. Habituation, another possible explanation of CARE [28], proposes that exposure to VR reduces the user's response to sickness-inducing stimuli (e.g., graphics latency or visual motion) through learning. If the sickness-inducing stimuli are common across both games, then the habituation account predicts similar generalization of CARE for both games in the Repeated Exposure conditions. Based on these predictions, the Day 4 results support the habituation account, because cybersickness is lower in both Repeated Exposure conditions compared to the Single Exposure condition and did not significantly differ between the two Repeated Exposure conditions. However, because this study was not designed to serve as a direct theory test, these theoretical interpretations should be viewed as suggestive rather than conclusive.

It is not clear why a related study [16] did not find generalization of CARE between a roller coaster experience and a climbing game. There are many differences between that study and the current study, including differences in the specific stimuli/games used, whether locomotion was passive or active, and the design of the experimental conditions. Additionally, the effect sizes in the current study when comparing Day 4 cybersickness to that in the Single Exposure condition are considered small-to-medium. An effect of this size may not have been observable in previous studies due to smaller sample sizes. More research is needed to identify the factors that moderate generalization of CARE.

Across conditions, several individual differences reliably predicted cybersickness. Consistent with prior work [8], [9], women reported higher sickness than men, and participants with greater visually induced motion sickness susceptibility experienced more severe symptoms. Exploratory moderation analyses identified a small number of gender-by-condition interactions, raising the possibility that the magnitude of CARE generalization may

[1]Meta does not provide exact data on individual game sales, but there are currently >1 k reviews of each game on the Meta Quest store, and the true number of users likely far exceeds the number of users who have reviewed the game.

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differ between men and women. It is possible that baseline susceptibility (which is higher among women) could interact with the rate and the ceiling of adaptation across repeated exposure. One interpretation of the present findings is that prior VR exposure reduces global sickness severity for both groups, while residual differences in susceptibility may remain more pronounced in some contexts, such as following exposure to Shadow Point. At the same time, these moderation effects were modest, inconsistent across dependent measures, and not part of a priori hypotheses, and they did not alter the overall pattern of CARE generalization. Accordingly, the present data do not support the adoption of gender-specific CARE protocols, but instead motivate larger, pre-registered studies that systematically vary exposure schedules and locomotion demands to test whether the time course, durability, or upper bounds of CARE differ as a function of gender.

From a practical standpoint, the present findings suggest that brief, structured prior exposure to VR can lead to benefits that extend to a novel application. For developers and practitioners, this implies that onboarding experiences or tutorials that provide short, repeated exposures across multiple sessions may be more effective than a single, extended introductory session. For example, applications deployed in training, education, or entertainment contexts could incorporate optional multi-day acclimation modes before users engage with more demanding content. For VR users, these results suggest that spacing early VR use across several short sessions may improve comfort when transitioning to new VR applications. At the same time, the present data do not establish a precise "dose" of exposure required for generalization, and the observed benefits were modest and context-dependent, highlighting the need for future work that systematically varies the number, duration, and spacing of exposure sessions to determine optimal schedules.

A limitation of the present study is the unequal sample sizes across conditions, with a larger Single Exposure group than either Repeated Exposure group. Because statistical power in between-group comparisons is constrained by the smallest group sizes, the relatively smaller repeated-exposure samples limit sensitivity for detecting small-to-moderate differences, particularly between the two Repeated Exposure conditions. To address this concern, the primary generalization analyses were complemented by a covariate-adjusted, gender-stratified bootstrap procedure that resampled from the Single Exposure pool to match the size and composition of each Repeated Exposure group. This sensitivity analysis reproduced the same pattern of results, suggesting that the primary conclusions are robust to group size imbalance.

Another limitation is the relatively high attrition rate in the Repeated Exposure conditions (36%). Although analyses in Section IV-B indicated no significant differences between completers and non-completers on key demographics or Day 1 cybersickness measures, these tests cannot fully rule out selection effects that emerge after initial exposure (e.g., differential withdrawal by participants who experience escalating symptoms across days). As a result, the observed generalization effects may overestimate CARE if dropout was systematically related to individual tolerance for repeated VR exposure.

Taken together, the present results demonstrate that prior VR exposure generalizes across distinct VR contexts, as evidenced by lower FMS ratings on Day 4 in both Repeated Exposure conditions compared to the Single Exposure condition. This pattern suggests that CARE may therefore serve as an effective mitigation approach for reducing sickness during a user's first experience with a new VR application, at least for subjective discomfort measured by FMS. At the same time, the weaker and less consistent effects observed for SSQ outcomes suggest that different symptom domains may generalize to different extents, consistent with the partial overlap between FMS and SSQ measures observed in this and prior work [12], [35]. Finally, the mixed within-game CARE trajectories, particularly in the Climb condition, likely reflect uncontrolled variation in task difficulty and motion intensity as participants progressed through the commercial games, highlighting a trade-off between ecological validity and experimental control.

VI. Future Directions

The finding that CARE generalizes to a distinct VR context is encouraging for its further development as a cybersickness mitigation tool. However, additional research is needed to establish optimal CARE protocols.

The present study used commercially available VR games to maximize ecological validity and reflect the mixed, naturalistic visual motion that typical VR users experience. Future research could complement this approach by systematically manipulating specific visual-motion components (e.g., isolated linear or rotational optic flow). Such designs would enable fine-grained investigation of which motion cues support or limit the generalization of CARE, and how reductions in sickness transfer across controlled motion profiles.

Beyond stimulus design, several additional questions warrant future investigation. One important question is whether CARE correlates with individual differences, such as those known to influence initial cybersickness [8], [9], [10], [12], [41]. The present exploratory analyses suggested that gender may modestly moderate generalization under some conditions, but these effects were not consistent across outcomes. Larger, pre-registered studies are needed to more definitively characterize how gender and other individual differences shape CARE and its generalization. Another issue is whether CARE is durable over time or whether it must be reestablished after periods without VR exposure. Finally, it would be valuable to identify ways to reduce the initial discomfort that can occur before the benefits of CARE accrue. One study [14] indicates that CARE still occurs when using FOV reduction to reduce cybersickness, but other mitigation strategies (e.g., reducing exposure time) should also be investigated.

VII. Conclusion

Repeated exposure to VR is known to cause CARE [14], [16], [21], which represents a potential solution for VR users to eventually experience VR without significant cybersickness. The results of the current study paint a hopeful picture, whereby CARE that occurs in one VR context generalizes to a new VR

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context. These conclusions should be interpreted in light of several limitations, including the use of commercial games with uncontrolled visual motion parameters, unequal group sizes and attrition in the repeated-exposure conditions, and the modest effect sizes observed for some outcomes, which together constrain the precision and generalizability of the present estimates of CARE and its generalization. If generalization of CARE occurs broadly, then VR users who have achieved significant reduction in cybersickness through repeated exposure may be able to explore new and diverse VR experiences without concern of returning to initial levels of sickness.

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